University Of Asia Pacific

Course Code: CSE 402

Course Title: Operating System Lab

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Problem Statement : FIFO_page_replacement

Request Queue = 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

capacity = 3

Source Code :

```
def fifo_page_replacement(pages, capacity):
    memory = []
    page_faults = 0

    for page in pages:
        if page not in memory:

            if len(memory) == capacity:
                memory.pop(0)

            memory.append(page)
            page_faults += 1

    return page_faults, memory


pages = [1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5]
capacity = 3

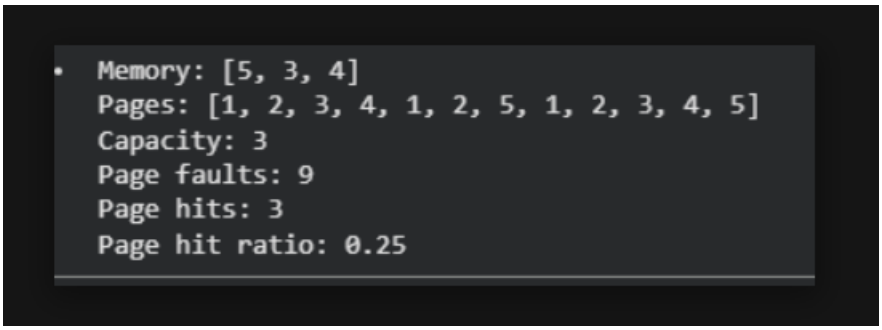
page_faults, memory = fifo_page_replacement(pages, capacity)

page_hits = len(pages) - page_faults
hit_ratio = page_hits / len(pages)

print("Memory:", memory)
print("Pages:", pages)
print("Capacity:", capacity)
print("Page faults:", page_faults)
```

```
print("Page hits:", page_hits)
print("Page hit ratio:", hit_ratio)
```

Output :



Discussion:

This program implements the FIFO (First-In, First-Out) Page Replacement Algorithm in Operating Systems. It checks each page in the given reference string. If a page is not present in memory, a page fault occurs. When memory is full, the oldest page is removed and replaced with the new page. The program also calculates the total page faults, page hits, and page hit ratio.

GitHub Link : https://github.com/JoynoveMim/Operating-System.git

Result:

The program was executed using a page reference string of 12 pages and a memory capacity of 3 frames. The final memory contains pages 4, 5, and 3. A total of 9 page faults and 3 page hits occurred. The page hit ratio is 0.25, which means 25% of the page requests were found in memory. The result shows how the FIFO algorithm replaces the oldest page when memory becomes full.

Conclusion:

The FIFO page replacement algorithm replaces the oldest page when memory is full. This program successfully calculates page faults, page hits, and the page hit ratio. It is simple to implement, but it may replace a frequently used page because it considers only the arrival order of pages.